

20 Ans: A

Option A: $V = 0$ as voltmeter measuring the same potential across the two ends

Option B: $V = 12\text{ V}$ as voltmeter is in parallel to 12 V battery

Option C: As V is placed in series with significant much higher resistance than the other resistors, it will take all the 12 V by potential divider

Option D: $V = 12\text{ V}$ as voltmeter is in parallel to 12 V battery

A significant number of candidates did not appear to understand that a voltmeter has a very high resistance and as a consequence selected option C.

21 Ans: B